

Exercise 12.2

1. Find the arithmetic mean in each of the following:

(i) 4, 6, 10, 12, 15, 20, 25, 28, 30.

X
4
6
10
12
15
20
25
28
30
$\Sigma X = 150$

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{150}{9} \\ \bar{X} &= 16.67\end{aligned}$$

(ii) 12, 18, 19, 0, -19, -18, -12

X
12
18
19
0
-19
-18
-12
$\Sigma X = 0$

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{0}{7} \\ \bar{X} &= 0\end{aligned}$$

(iii) 6.5, 11, 12.3, 9, 8.1, 16, 18, 20.5, 25

X
6.5
11
12.3
9
8.1
16
18
20.5
25
$\Sigma X = 126.4$

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{126.4}{9} \\ \bar{X} &= 14.044\end{aligned}$$

(iv) 8, 10, 12, 14, 16, 20, 22

X
8
10
12
14
16
20
22

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{102}{7} \\ \bar{X} &= 14.57\end{aligned}$$

$$\Sigma X = 102$$

2. Following are the heights in (inches) of 12 students. Find the median height.

55, 53, 54, 58, 60, 61, 62, 56, 57, 52, 51, 63.

By arranging

51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 62, 63

Since number of observations is even i.e. $n = 12$

$$\begin{aligned}\tilde{X} &= \frac{1}{2} \left[\left(\frac{n}{2} \right)^{th} \text{ observation} + \left(\frac{n+2}{2} \right)^{th} \text{ observation} \right] \\ \tilde{X} &= \frac{1}{2} \left[\left(\frac{12}{2} \right)^{th} \text{ observation} + \left(\frac{12+2}{2} \right)^{th} \text{ observation} \right] \\ \tilde{X} &= \frac{1}{2} [(6)^{th} \text{ observation} + (7)^{th} \text{ observation}] \\ \tilde{X} &= \frac{1}{2} [56 + 57] \\ \tilde{X} &= \frac{113}{2} \\ \tilde{X} &= 56.5 \text{ inches}\end{aligned}$$

3. Following are the earnings (in Rs.) of ten workers: 88, 70, 72, 125, 115, 95, 81, 90, 95, 90.

Calculate (i) Arithmetic Mean

X
88
70
72
125
115
95
81
90
95
90
$\Sigma X = 921$

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{921}{10} \\ \bar{X} &= 92.1\end{aligned}$$

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(ii) Median

By arranging

70, 72, 81, 88, 90, 90, 95, 95, 115, 125

Since number of observations is even i.e. $n = 10$

$$\begin{aligned}\tilde{X} &= \frac{1}{2} \left[\left(\frac{n}{2} \right)^{th} \text{ observation} + \left(\frac{n+2}{2} \right)^{th} \text{ observation} \right] \\ \tilde{X} &= \frac{1}{2} \left[\left(\frac{10}{2} \right)^{th} \text{ observation} + \left(\frac{10+2}{2} \right)^{th} \text{ observation} \right] \\ \tilde{X} &= \frac{1}{2} [(5)^{th} \text{ observation} + (6)^{th} \text{ observation}] \\ \tilde{X} &= \frac{1}{2} [90 + 90] \\ \tilde{X} &= \frac{180}{2} \\ \tilde{X} &= 90\end{aligned}$$

(iii) Mode

$$\begin{aligned}\text{Mode} &= \hat{X} = \text{the most frequent observation} \\ \hat{X} &= 90, 95\end{aligned}$$

Note: In the data:

- 90 appears 2 times
- 95 appears 2 times
- Others appear once.

4. The Marks obtained by the students in the subject of English are given below.

Marks obtained	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39
Frequency	9	18	35	17	5

Find: (i) Arithmetic mean of their marks by direct and short formula.

Arithmetic mean by direct method:

Class limits	f	X	fX
15 – 19	9	17	153
20 – 24	18	22	396
25 – 29	35	27	945
30 – 34	17	32	544
35 – 39	5	37	185
	$\Sigma f = 84$		$\Sigma fX = 2223$

$$\bar{X} = \frac{\Sigma fX}{\Sigma f}$$

$$\bar{X} = \frac{2223}{84}$$

$$\bar{X} = 26.46$$

Arithmetic mean by short formula: Let $A = 27$

Class limits	f	X	$D = X - A$ $D = X - 27$	fD
15 – 19	9	17	-10	-90
20 – 24	18	22	-5	-90
25 – 29	35	27	0	0
30 – 34	17	32	5	85
35 – 39	5	37	10	50
	$\Sigma f = 84$			$\Sigma fD = -45$

$$\bar{X} = A + \frac{\Sigma fD}{\Sigma f}$$

$$\bar{X} = 27 + \frac{-45}{84}$$

$$\bar{X} = 27 - 0.5357$$

$$\bar{X} = 26.46$$

(ii) Median of their marks.

Class	f	Class Boundaries (C. B)	Cumulative Frequency (c. f)	
15 – 19	9	14.5 – 19.5	9	
20 – 24	18	19.5 – 24.5	9 + 18 = 27	
25 – 29	35	24.5 – 29.5	27 + 35 = 62	→ Median Class
30 – 34	17	29.5 – 34.5	62 + 17 = 79	
35 – 39	5	34.5 – 39.5	79 + 5 = 84	

Here $n = 84$, So $\frac{n}{2} = \frac{84}{2} = 42$

Hence 42th item lie in 24.5 – 29.5

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\begin{aligned}\tilde{X} &= 24.5 + \frac{5}{35} \left(\frac{84}{2} - 27 \right) \\ \tilde{X} &= 24.5 + (0.1428)(42 - 27) \\ \tilde{X} &= 24.5 + (0.1428)(15) \\ \tilde{X} &= 24.5 + 2.142 \\ \tilde{X} &= 26.642\end{aligned}$$

5. Given below is a frequency distribution.

Class Interval	5 – 9	10 – 14	15 – 19	20 – 24	25 – 29
Frequency	1	8	18	11	2

Find the mode of the frequency distribution.

Class Interval	Frequency (f)	Class Boundaries($C.B$)	
5 – 9	1	4.5 – 9.5	
10 – 14	8	9.5 – 14.5	
15 – 19	18	14.5 – 19.5	→ Modal Class
20 – 24	11	19.5 – 24.5	
25 – 29	2	24.5 – 29.5	

Modal Class = the most frequent observation
 $= 14.5 - 19.5$

$$\begin{aligned}\text{Mode} &= l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h \\ \hat{X} &= 14.5 + \frac{(18 - 8)}{(18 - 8) + (18 - 11)} \times 5 \\ \hat{X} &= 14.5 + \frac{10}{50} \times 5 \\ \hat{X} &= 14.5 + \frac{10}{17} \\ \hat{X} &= 14.5 + 2.94 \\ \hat{X} &= 17.47\end{aligned}$$

6. Ten boys work on a petrol pump station. They get weekly wages as follows: Wages (in Rs.) 4250, 4350, 4400, 4250, 4350, 4410, 4500, 4300, 4500, 4390. Find the arithmetic mean by short formula, median and mode of their wages.

Arithmetic mean by short formula: Let $A = 4350$

Wages (X)	$D = X - 4350$
4250	-100
4250	-100
4300	-50
4350	0
4350	0
4390	40
4400	50
4410	60
4500	150
4500	150
	$\sum D = 200$

$$\begin{aligned}\bar{X} &= A + \frac{\sum D}{n} \\ \bar{X} &= 4350 + \frac{200}{10}\end{aligned}$$

$$\bar{X} = 4350 + 20$$

$$\bar{X} = \mathbf{4370 \text{ Rs.}}$$

Median: By arranging

4250, 4250, 4300, 4350, 4350, 4390, 4400, 4410, 4500, 4500

Since number of observations is even i.e. $n = 10$

$$\tilde{X} = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{th} \text{ observation} + \left(\frac{n+2}{2} \right)^{th} \text{ observation} \right]$$

$$\tilde{X} = \frac{1}{2} \left[\left(\frac{10}{2} \right)^{th} \text{ observation} + \left(\frac{10+2}{2} \right)^{th} \text{ observation} \right]$$

$$\tilde{X} = \frac{1}{2} [(5)^{th} \text{ observation} + (6)^{th} \text{ observation}]$$

$$\tilde{X} = \frac{1}{2} [4350 + 4390]$$

$$\tilde{X} = \frac{8740}{2}$$

$$\tilde{X} = \mathbf{4370 \text{ Rs.}}$$

Mode:

$$\text{Mode} = \hat{X} = \text{the most frequent observation}$$

$$\hat{X} = \mathbf{4250, 4350 \text{ and } 4500}$$

Note: From data:

- 4250 occurs 2 times
- 4350 occurs 2 times
- 4500 occurs 2 times
- Others occur once.

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7. The arithmetic mean of 45 numbers is 80. Find their sum.

$$\bar{X} = 80$$

$$n = 45$$

$$\Sigma X = ?$$

By using formula

$$\bar{X} = \frac{\Sigma X}{n}$$

$$80 = \frac{\Sigma X}{45}$$

$$80 \times 45 = \Sigma X$$

$$3600 = \Sigma X$$

$$\Sigma X = \mathbf{3600}$$

8. Five numbers are 1, 4, 0, 7, 9. Find their mean, median and mode.

Mean:

X
1
4
0
7
9
$\Sigma X = 21$

$$\bar{X} = \frac{\Sigma X}{n}$$

$$\bar{X} = \frac{21}{5}$$

$$\bar{X} = \mathbf{4.2}$$

Median: By arranging

0, 1, 4, 7, 9

Since number of observations is odd i.e. $n = 5$

$$\tilde{X} = \left(\frac{n+1}{2}\right)^{th} \text{ observation}$$

$$\tilde{X} = \left(\frac{5+1}{2}\right)^{th} \text{ observation}$$

$$\tilde{X} = 3^{rd} \text{ observation}$$

$$\tilde{X} = 4$$

Mode:

$\hat{X} = \text{the most frequent observation}$

$\hat{X} = \text{no mode} \quad \because \text{no repetition}$

9. A set of data contains the values as 148, 145, 160, 157, 156, 160. Show that $\text{Mode} > \text{Median} > \text{Mean}$.

Mean:

X
148
145
160
157
156
160
$\Sigma X = 926$

$$\bar{X} = \frac{\Sigma X}{n}$$

$$\bar{X} = \frac{926}{6}$$

$$\bar{X} = 154.33$$

Median: By arranging

145, 148, 156, 157, 160, 160

Since number of observations is even i.e. $n = 6$

$$\tilde{X} = \frac{1}{2} \left[\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{n+2}{2}\right)^{th} \text{ observation} \right]$$

$$\tilde{X} = \frac{1}{2} \left[\left(\frac{6}{2}\right)^{th} \text{ observation} + \left(\frac{6+2}{2}\right)^{th} \text{ observation} \right]$$

$$\tilde{X} = \frac{1}{2} [(3)^{rd} \text{ observation} + (4)^{th} \text{ observation}]$$

$$\tilde{X} = \frac{1}{2} [156 + 157]$$

$$\tilde{X} = \frac{313}{2}$$

$$\tilde{X} = 156.5$$

Mode:

$\hat{X} = \text{the most frequent observation}$

$\hat{X} = 160$

So, $\text{Mode} (160) > \text{Median} (156.5) > \text{Mean} (154.3)$

10. The monthly attendance of 10 students for their lunch in the hostel is recorded as: 21, 15, 16, 18, 14, 17, 15, 12, 13, 11. Find the median and mode of the attendance. Also find the mean if $D = A - 20$.

Here $D = A - 20$. So Assumed Mean = 20

Attendance (A)	$D = A - 20$
11	-9
12	-8
13	-7
14	-6

15	-5
15	-5
16	-4
17	-3
18	-2
21	+1
	$\Sigma D = -48$

$$\bar{X} = A + \frac{\Sigma D}{n}$$

$$\bar{X} = 20 + \frac{-48}{10}$$

$$\bar{X} = 20 - 4.8$$

$$\bar{X} = 15.2$$

11. On a prize distribution day, 50 students brought pocket money as under:

Rupees	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
Frequency (f)	12	9	18	7	4

(i) Find the median and mode of the above data.

Median:

Class	f	Class Boundaries (C. B)	Cumulative Frequency (c. f)	
5 – 10	12	5 – 10	12	
10 – 15	9	10 – 15	21	
15 – 20	18	15 – 20	39	→ Median Class
20 – 25	7	20 – 25	46	
25 – 30	4	25 – 30	50	

Here $n = 50$, So $\frac{n}{2} = \frac{50}{2} = 25$

Hence 25th item lie in 15 – 20

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\tilde{X} = 15 + \frac{5}{18} \left(\frac{50}{2} - 21 \right)$$

$$\tilde{X} = 15 + (0.2778)(25 - 21)$$

$$\tilde{X} = 15 + (0.2778)(4)$$

$$\tilde{X} = 15 + 1.1112$$

$$\tilde{X} = 16.1112$$

Note: In grouped data, if classes are continuous (e.g. 5 – 10, 10 – 15), class width h is taken directly, but if classes are inclusive (e.g. 15 – 19, 20 – 24), we first use class boundaries to remove gaps, and then find h .

Mode:

Class Interval	Frequency (f)	Class Boundaries (C. B)	
5 – 10	12	5 – 10	
10 – 15	9	10 – 15	
15 – 20	18	15 – 20	→ Modal Class
20 – 25	7	20 – 25	
25 – 30	4	25 – 30	

Modal Class = the most frequent observation
= 15 – 20

$$\text{Mode} = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h$$

$$\hat{X} = 15 + \frac{(18 - 9)}{(18 - 9) + (18 - 7)} \times 5$$

$$\hat{X} = 15 + \frac{9}{9 + 11} \times 5$$

$$\hat{X} = 15 + \frac{45}{20}$$

$$\hat{X} = 15 + 2.25$$

$$\hat{X} = 17.25$$

(ii) Find the arithmetic mean of the data given above using coding method.

Let $A = 17.5$, class width $h = 5$

Class	f	Midpoint (X)	$U = \frac{X - A}{h}$ $U = \frac{X - 17.5}{5}$	fU
5 – 10	12	7.5	-2	-24
10 – 15	9	12.5	-1	-9
15 – 20	18	17.5	0	0
20 – 25	7	22.5	1	7
25 – 30	4	27.5	2	8
	$\Sigma f = 50$			$\Sigma fU = -18$

$$\bar{X} = A + \frac{\Sigma fU}{\Sigma f} \times h$$

$$\bar{X} = 17.5 + \frac{-18}{50} \times 5$$

$$\bar{X} = 17.5 - \frac{90}{50}$$

$$\bar{X} = 17.5 - 1.5$$

$$\bar{X} = 15.7$$

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12. The arithmetic mean of the ages of 20 boys is 13 years, 4 months and 5 days. Find the sum of their ages. If one of the boys is of age exactly 15 years. What is the average age of the remaining boys?

$$\text{No. of boys} = n = 20$$

$$\text{Arithmetic mean} = \bar{X} = 13 \text{ years, 4 months and 5 days}$$

$$\bar{X} = 13 \text{ years} + 4 \text{ months} + 5 \text{ days}$$

$$\bar{X} = \left(13 + \frac{4}{12} + \frac{5}{365} \right) \text{ years}$$

$$\bar{X} = (13 + 0.3333 + 0.0137) \text{ years}$$

$$\bar{X} = 13.374 \text{ years}$$

$$\text{Sum of ages} = \Sigma X = ?$$

$$\text{Average age of the remaining boys} = ?$$

$$(\text{When age of one boy} = 15 \text{ years})$$

By using formula

$$\bar{X} = \frac{\Sigma X}{n}$$

$$13.374 = \frac{\Sigma X}{20}$$

$$13.374 \times 20 = \Sigma X$$

$$266.94 = \Sigma X$$

$$\Sigma X = 266.94 \text{ years}$$

$$\Sigma X = 266 \text{ yrs.} + 0.94 \text{ yrs.}$$

$$\Sigma X = 266 \text{ yrs.} + (0.94 \times 12) \text{ months}$$

$$\begin{aligned}\Sigma X &= 266 \text{ yrs.} + 11.28 \text{ months} \\ \Sigma X &= 266 \text{ yrs.} + 11 \text{ months} + 0.28 \text{ months} \\ \Sigma X &= 266 \text{ yrs.} + 11 \text{ months} + (0.28 \times 31) \text{ days} \\ \Sigma X &= 266 \text{ yrs.} + 11 \text{ months} + 8.68 \text{ days} \\ \Sigma X &= \mathbf{266 \text{ years, 11 months and 9 days}}\end{aligned}$$

Now

$$\begin{aligned}\text{Age of remaining 19 boys} &= 266.97 - 15 \\ \Sigma X &= \mathbf{251.94}\end{aligned}$$

Now again by using formula

$$\begin{aligned}\bar{X} &= \frac{\Sigma X}{n} \\ \bar{X} &= \frac{251.94}{19} \\ \bar{X} &= 13.26 \text{ years} \\ \bar{X} &= 13 \text{ yrs.} + 0.26 \text{ yrs.} \\ \bar{X} &= 13 \text{ yrs.} + (0.26 \times 12) \text{ months} \\ \bar{X} &= 13 \text{ yrs.} + 3.12 \text{ months} \\ \bar{X} &= 13 \text{ yrs.} + 3 \text{ months} + 0.12 \text{ months} \\ \bar{X} &= 13 \text{ yrs.} + 3 \text{ months} + (0.12 \times 30) \text{ days} \\ \bar{X} &= 13 \text{ yrs.} + 3 \text{ months} + 3.6 \text{ days} \\ \bar{X} &= \mathbf{13 \text{ years, 3 months and 4 days}}\end{aligned}$$

13. Calculate the arithmetic mean from the following information:

(i) If $D = X - 140$, $\Sigma D = 500$ and $n = 10$

Give that

$$\begin{aligned}D &= X - 140 \\ \Sigma D &= 500 \\ n &= 10 \\ \bar{X} &= ?\end{aligned}$$

As $D = X - 140$. By comparing with $D = X - A$, $\Rightarrow A = 140$

By using formula

$$\begin{aligned}\bar{X} &= A + \frac{\Sigma D}{n} \\ \bar{X} &= 140 + \frac{500}{10} \\ \bar{X} &= 140 + 50 \\ \bar{X} &= \mathbf{90}\end{aligned}$$

(ii) If $U = \frac{X-130}{6}$, $\Sigma U = -150$ and $n = 15$

Give that

$$\begin{aligned}U &= \frac{X - 130}{6} \\ \Sigma U &= -150 \\ n &= 15 \\ \bar{X} &= ?\end{aligned}$$

As $U = \frac{X-130}{6}$. By comparing with $U = \frac{X-A}{h}$, $\Rightarrow A = 130$ and $h = 6$

By using formula

$$\begin{aligned}\bar{X} &= A + \frac{\Sigma U}{n} \times h \\ \bar{X} &= 130 + \frac{-150}{15} \times 6\end{aligned}$$

$$\bar{X} = 130 - \frac{900}{15}$$

$$\bar{X} = 130 - 60$$

$$\bar{X} = 70$$

(iii) If $D = X - 25$, $\sum fD = 300$ and $\sum f = 20$

Give that

$$D = X - 25$$

$$\sum fD = 300$$

$$\sum f = 20$$

$$\bar{X} = ?$$

As $D = X - 25$. By comparing with $D = X - A$, $\Rightarrow A = 25$

By using formula

$$\bar{X} = A + \frac{\sum fD}{\sum f}$$

$$\bar{X} = 25 + \frac{300}{20}$$

$$\bar{X} = 25 + 15$$

$$\bar{X} = 40$$

(iv) If $U = \frac{X-120}{5}$, $\sum fU = 60$ and $\sum f = 100$

Give that

$$U = \frac{X - 120}{5}$$

$$\sum fU = 60$$

$$\sum f = 100$$

$$\bar{X} = ?$$

As $U = \frac{X-120}{5}$. By comparing with $U = \frac{X-A}{h}$, $\Rightarrow A = 120$ and $h = 5$

By using formula

$$\bar{X} = A + \frac{\sum fU}{\sum f} \times h$$

$$\bar{X} = 120 + \frac{60}{100} \times 5$$

$$\bar{X} = 120 + \frac{300}{100}$$

$$\bar{X} = 120 + 3$$

$$\bar{X} = 123$$

14. The three children Haris, Maham and Minal made the following scores in a game conducted by a group of teachers in the school.

Haris scores	50	55	70	85	90
Maham scores	75	60	60	45	53
Minal scores	80	77	66	42	48

It is decided that the candidate who gets the highest average score will be awarded rupees 1000. Who will get the awarded amount?

Haris (X_H)	Maham (X_M)	Minal (X_N)
50	75	80
55	60	77
70	60	66
85	45	42

90	53	48
$\Sigma X_H = 350$	$\Sigma X_M = 293$	$\Sigma X_N = 313$

$$\begin{aligned}\text{Average score of Haris} &= \bar{X}_H = \frac{\Sigma X_H}{n} \\ &= \frac{350}{5} \\ \bar{X}_H &= 70\end{aligned}$$

$$\begin{aligned}\text{Average score of Maham} &= \bar{X}_M = \frac{\Sigma X_M}{n} \\ &= \frac{293}{5} \\ \bar{X}_M &= 58.6\end{aligned}$$

$$\begin{aligned}\text{Average score of Minal} &= \bar{X}_N = \frac{\Sigma X_N}{n} \\ &= \frac{313}{5} \\ \bar{X}_N &= 62.6\end{aligned}$$

Since Haris has the highest average score (70). He will get the awarded amount of Rs. 1000.

15. Given below is a frequency distribution derived by making a substitution as $D = X - 20$.

D	-6	-4	-2	0	2	4	6
f	1	3	6	20	26	12	2

Calculate the arithmetic mean.

As $D = X - 20$. By comparing with $D = X - A, \Rightarrow A = 20$

D	f	fD
-6	1	-6
-4	3	-12
-2	6	-12
0	20	0
2	26	52
4	12	48
6	2	12
	$\Sigma f = 70$	$\Sigma fD = 82$

By using formula

$$\begin{aligned}\bar{X} &= A + \frac{\Sigma fD}{\Sigma f} \\ \bar{X} &= 20 + \frac{82}{70} \\ \bar{X} &= 20 + 1.17 \\ \bar{X} &= 21.17\end{aligned}$$

16. Being partners Hafsa and Fatima took part in a quiz programme. They made the following number of points 45, 51, 58, 61, 74, 48, 46 and 50. Compute the average number of points using deviation $D = X - 58$.

As $D = X - 58$.

By comparing with $D = X - A, \Rightarrow A = 58$

X	$D = X - 58$
45	-13
51	-7

58	0
61	3
74	16
48	-10
46	-12
50	-8
	$\Sigma D = -31$

By using formula

$$\bar{X} = A + \frac{\Sigma D}{n}$$

$$\bar{X} = 58 + \frac{-31}{8}$$

$$\bar{X} = 58 - 3.875$$

$$\bar{X} = 54.125$$

17. A person purchased the following food items:

Food item	Quantity (in Kg)	Cost per Kg (in Rs.)
Rice	10	96
Flour	12	48
Ghee	4	190
Sugar	3	49
Mutton	2	650

What is the weighted mean of cost of food items per kg?

Food item	Quantity (W)	Cost per Kg (X)	WX
Rice	10	96	960
Flour	12	48	576
Ghee	4	190	760
Sugar	3	49	147
Mutton	2	650	1300
	$\Sigma W = 31$		$\Sigma WX = 3743$

$$\bar{X}_w = \frac{\Sigma WX}{\Sigma W}$$

$$\bar{X}_w = \frac{3743}{31}$$

$$\bar{X}_w = 120.74$$

18. For the following data,

Item	Quantity	Cost of item (in thousands)
Washing Machine	5	35
Heater	3	5
Stove	2	13
Dispenser	6	18

find the weighted mean.

Item	Quantity (W)	Cost in thousands (X)	WX
Washing Machine	5	35	175
Heater	3	5	15
Stove	2	13	26
Dispenser	6	18	108
	$\Sigma W = 16$		$\Sigma WX = 324$

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$

$$\bar{X}_w = \frac{324}{16}$$

$$\bar{X}_w = 20.25 \text{ thousands}$$

19. A company is planning its next year marketing budget across five years: yearly budgets (in million) are: 5, 7, 8, 6, 7. Find the average budget for the next year.

Budget (X)
5
7
8
6
7
$\sum X = 33$

$$\bar{X} = \frac{\sum X}{n}$$

$$\bar{X} = \frac{33}{5}$$

$$\bar{X} = 6.6 \text{ million}$$

20. Ahmad obtained the following marks in a certain examination.

Urdu	English	Science	Math	Islamiyat	Computer
78	65	80	90	85	72

Find the weighted mean if weights 5, 4, 2, 3, 2, 4 respectively are allotted to the subjects.

Subject	Weight (W)	Marks (X)	WX
Urdu	5	78	390
English	4	65	260
Science	2	80	160
Math	3	90	270
Islamiyat	2	85	170
Computer	4	72	288
	$\sum W = 20$		$\sum WX = 1538$

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$

$$\bar{X}_w = \frac{1538}{20}$$

$$\bar{X}_w = 76.9$$